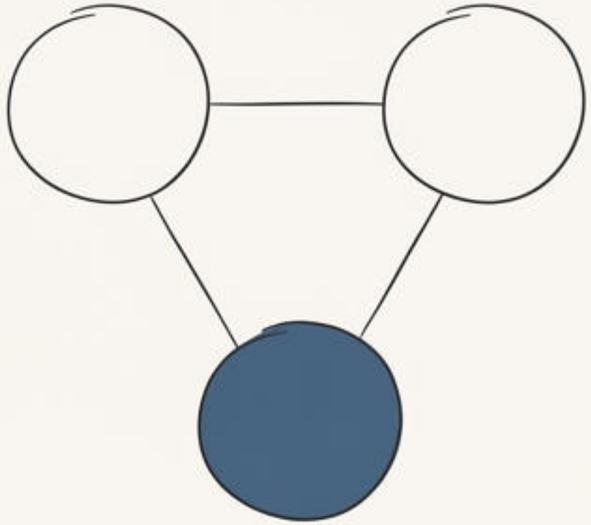
Classical Conditioning

An Introduction to **Pavlovian** and **Respondent Conditioning**



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Beyond Simple Reflexes

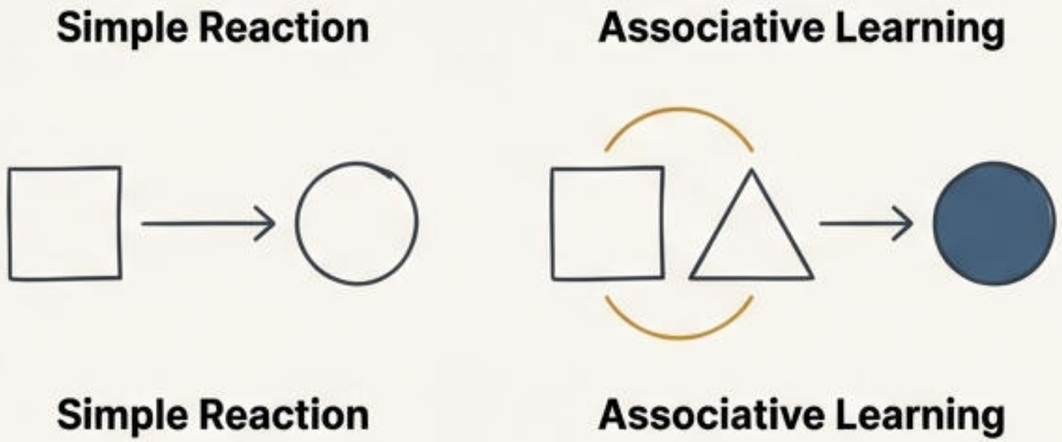
Previously, we explored how responses to a single, repeated stimulus can change through processes like habituation, sensitization, and opponent-processes. These are fundamental but limited forms of adaptation.

Anticipating a Complex World

In an environment filled with a vast array of stimuli, survival often depends on more than just reacting. It requires the ability to:

- **Anticipate** whether an event is about to occur.
- **Recognize** whether certain events are meaningfully **related** to other events.

The ability to relate one event to another allows an organism to better anticipate the future, thereby greatly facilitating its chances of survival. For instance, it is highly adaptive to associate the pain of a wasp sting with the sight and sound of the wasp.



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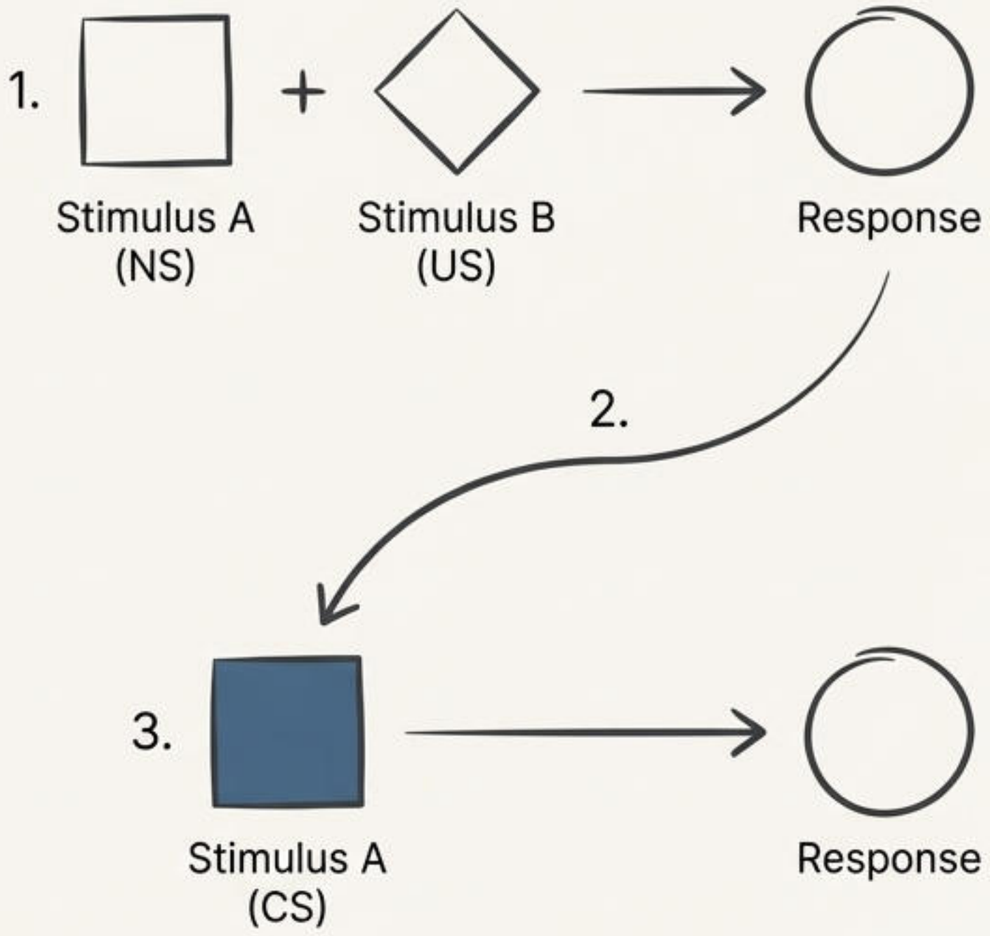
A Fundamental Process of Association

Classical conditioning is a process in which one stimulus that does not elicit a certain response is associated with a second stimulus that does; as a result, the first stimulus also comes to elicit a response.

Alternative Terminology

- **Pavlovian Conditioning:** Named after Ivan Pavlov, who first systematically investigated the principles of this learning process.
- **Respondent Conditioning:** This term is often preferred by behavior analysts who follow B.F. **Skinner's approach**. The elicited behaviors (like salivation or fear) are referred to as **respondent behaviors** or simply **respondents**.



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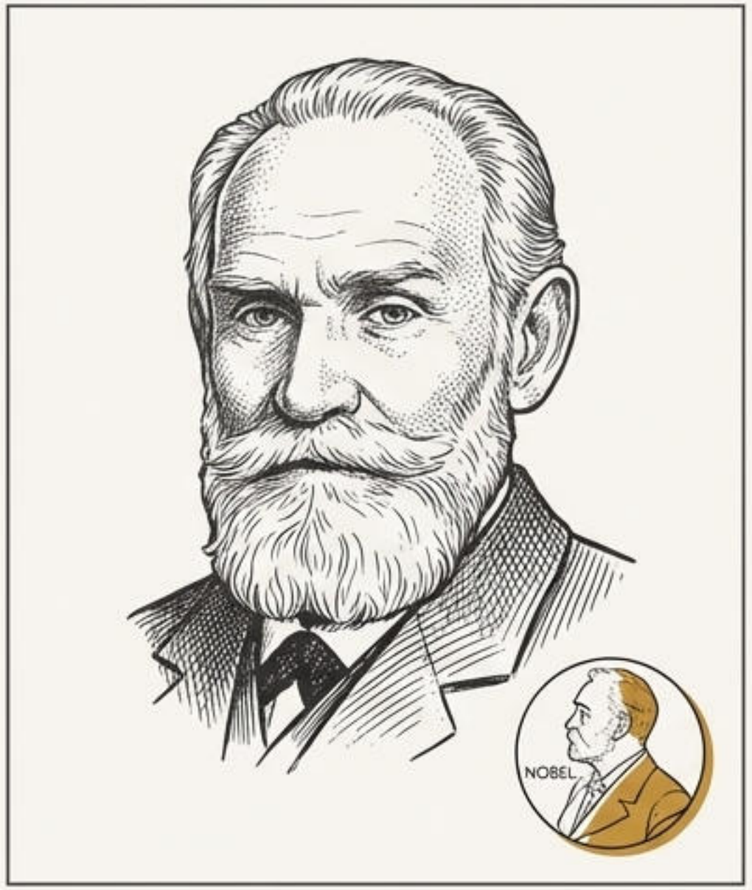
The Discovery of Classical Conditioning

Ivan P. Pavlov (1849–1936): A Foundation in Physiology



Ivan Pavlov, a renowned Russian physiologist, is credited with the first systematic investigations into **classical conditioning**. His primary research, beginning in the late 1800s, was on the physiology of digestive secretions secretions and their neural control. His contributions were so significant that he was awarded the Nobel Prize for this work.

As part of this research, Pavlov investigated salivation, the initial step in the digestive process. He was aware that salivation could be initiated by "psychic factors," such as the sight of food, but he was surprised by the lawfulness of these responses.

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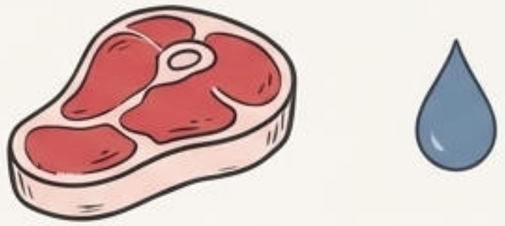
The Anomaly of “Psychic Secretions”

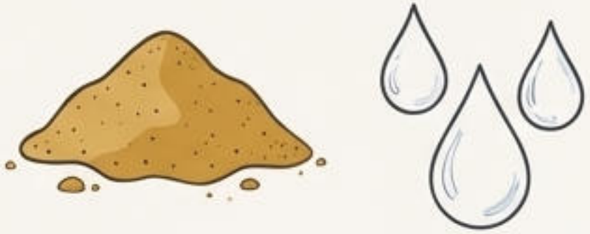
Pavlov noted that different substances affected both the quantity and quality of saliva produced, even when the dogs were merely shown the substances.

- **Moist, edible food (meat):** Elicited a small amount of slimy saliva.
- **Dry, inedible substance (sand):** Elicited a large amount of watery saliva (to facilitate ejection).

Subsequent research confirmed these psychic secretions exhibited a great deal of regularity and lawfulness. Fascinated by this, Pavlov began to devote more and more resources to their investigation. By 1907, what would become known as classical conditioning had become the sole focus of his research.



Small, slimy saliva



Large, watery saliva

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The Scientist and the Man

Pavlov was the epitome of the devoted scientist.

- **Disciplined:** He was a tough taskmaster, famously refusing an assistant's excuse for lateness due to revolutionary battles in the streets.
- **Principled:** He was **openly critical** of the Soviet regime's denial of basic rights and religious freedoms.
- **Humane:** He showed great concern for the welfare of his dogs, devising surgical procedures that allowed for accurate observation while minimizing discomfort and ensuring full postoperative recovery.

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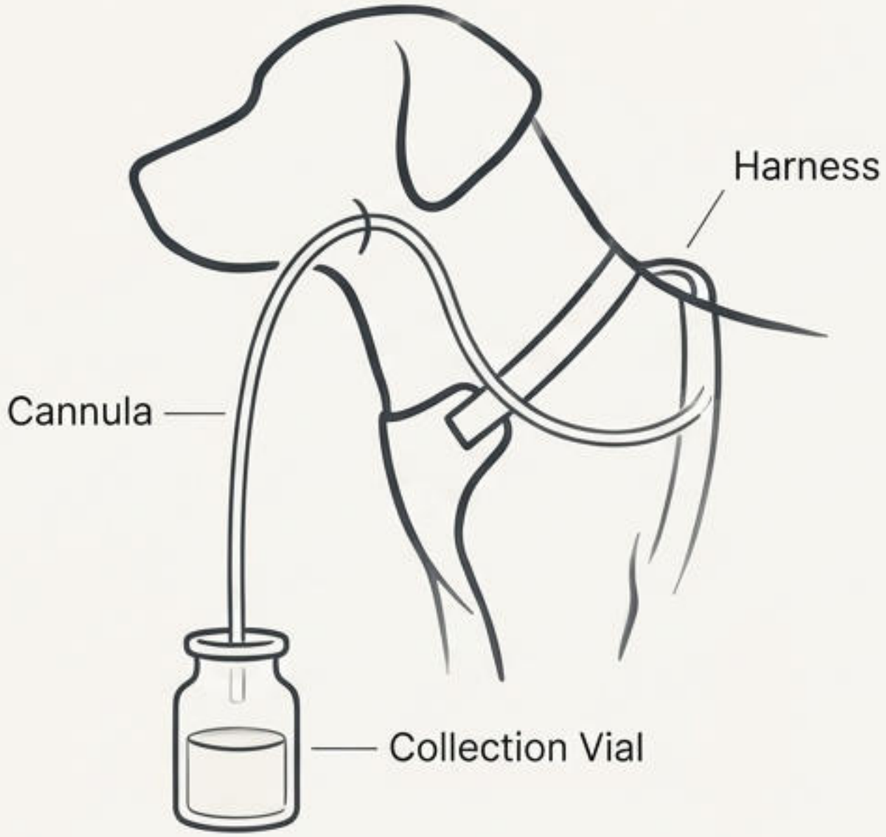
The Basic Procedure and Definitions

Pavlov's Experimental Apparatus

To study salivation systematically, Pavlov developed a controlled experimental setup.

- **Harness:** The dog was restrained in a harness to keep it still. The dogs habituated to this apparatus readily.
- **Cannula:** A tube was inserted into an incision in the dog's cheek.
- **Measurement:** Saliva would run down the tube into a container where it could be precisely measured.

This setup allowed for the objective measurement of the salivary response to various stimuli.



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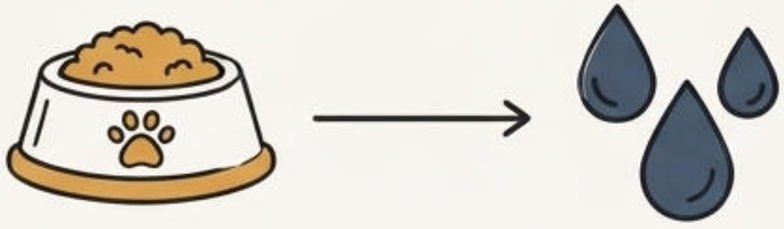
Phase 1: Before Conditioning

Before any learning occurs, we identify the existing stimulus-response relationships.

- **Unconditioned Stimulus (US):** A stimulus that naturally elicits a response without prior training. For example, **food**.
- **Unconditioned Response (UR):** The response that is naturally elicited by the US. For example, **salivation**. This relationship is innate or unlearned.
- **Neutral Stimulus (NS):** A stimulus that, prior to conditioning, does not elicit the response of interest. For example, the **sound of a metronome**.

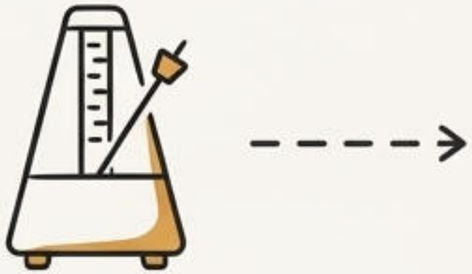
Schematic Representation

Food (US) ⟶ Salivation (UR)



US UR

Metronome (NS) ⟶ No Salivation



NS No Salivation

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Phase 2: During Conditioning

The conditioning phase involves repeatedly pairing the neutral stimulus with the unconditioned stimulus.

- The sound of the metronome **(NS)** is presented just before the food **(US)**.
- The food **(US)** continues to elicit the unconditioned response of salivation **(UR)**.
- Each pairing of the NS and the US is called a **conditioning trial**. Several trials are often needed for the NS to become established as a CS.

(Note: During this phase, the metronome can be labeled either an NS or a CS, because it begins as an NS and then becomes a CS.)

Schematic Representation:

Metronome (NS / CS) + Food (US) ⟶ Salivation (UR)

(NS / CS)

(US)

(UR)

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Phase 3: After Conditioning

After a sufficient number of conditioning trials, a new stimulus-response relationship has been formed.

- **Conditioned Stimulus (CS):** An initially neutral stimulus that, after association with an unconditioned stimulus, comes to elicit a response. The **sound of the metronome** is now a CS.
- **Conditioned Response (CR):** The response, often similar to the UR, that is elicited by the CS. **Salivation** in response to the metronome is the CR.

The metronome now elicits salivation because it has been associated with food.

Schematic Representation:

Metronome (CS) ⟶ Salivation (CR)

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Schematic of Classical Conditioning

1. Before Conditioning

Metronome (NS)

No Salivation

Food (US)

Salivation (UR)

2. During Conditioning

Metronome (NS)

+

Food (US)

Salivation (UR)

** (Repeated pairings establish an association)*

3. After Conditioning

Metronome (CS)

Salivation (CR)

** (The neutral stimulus has become a conditioned stimulus that now elicits a conditioned response.)*

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Application: Conditioning of Emotional Responses

Learning to Fear

Classical conditioning is a powerful mechanism for acquiring emotional responses. Consider a child who is bitten by a dog and subsequently develops a fear of dogs.

- The **bite** is an **Unconditioned Stimulus (US)** that elicits an **Unconditioned Response (UR)** of pain and fear.
- As a result of the bite being paired with the sight of the dog (previously a Neutral Stimulus), the dog becomes a **Conditioned Stimulus (CS)**.
- The sight of the dog now elicits a **Conditioned Response (CR)** of fear.

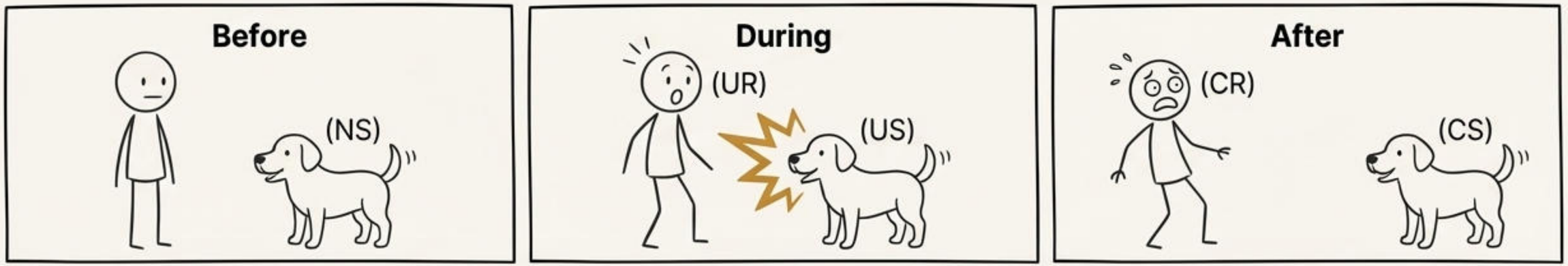
Schematic Representation:

Dog (NS) : Bite (US) ⟶ Fear (UR)

↓

Dog (CS) ⟶ Fear (CR)

Through this process, a previously neutral stimulus acquires significant behavioral importance.



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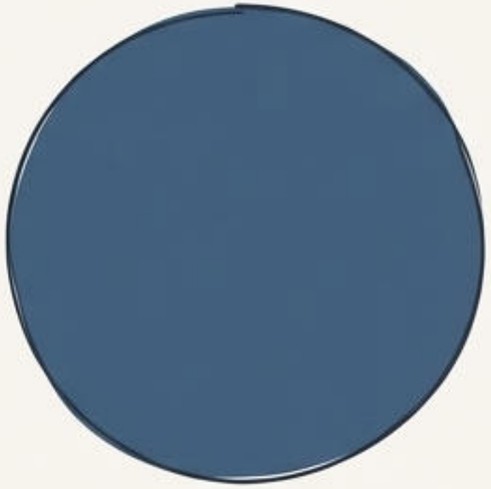
A Crucial Nuance: The CR is Not Identical to the UR

A common misconception is that the **conditioned response (CR)** is identical to the **unconditioned response (UR)**. While they are often similar, it is important to note they are never identical.

Key Distinctions:

- **Magnitude:** The **CR** is usually weaker or less intense than the **UR**. A dog will salivate less to the metronome (**CS**) than to the actual food (**US**).
- **Form:** The **CR** can sometimes be quite different from the **UR**. For example, in the dog-bite scenario:
  - The **UR** to the bite is primarily a reaction of **pain** ('Yeow!').
  - The **CR** to the subsequent sight of the dog is more accurately identified as **fear** (e.g., freezing, apprehension).

Understanding this distinction is crucial for a precise analysis of learned behaviors.



UR (Pain / Shock)

CR (Fear / Apprehension)

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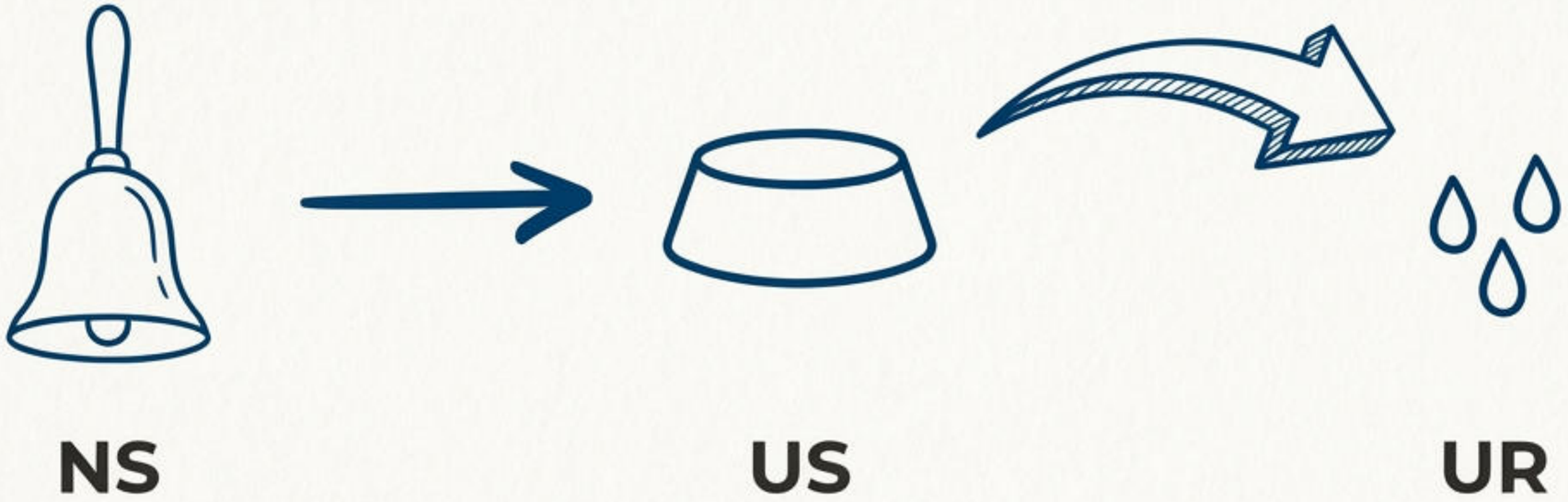
Mechanisms and Variations of Classical Conditioning

Appetitive and Aversive Conditioning



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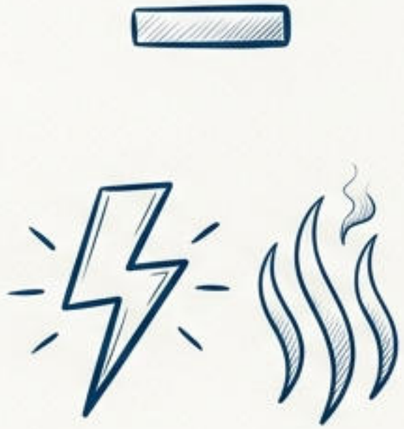
Appetitive and Aversive Conditioning

Most classical conditioning procedures can be classified into two primary categories based on the valence of the Unconditioned Stimulus (US): whether it is pleasant or unpleasant.

Appetitive Conditioning

In appetitive conditioning, the US is an event that an organism will generally approach or seek out. In mentalistic terms, it is something typically regarded as pleasant.

- Examples: Food (if the organism is hungry), water (if thirsty), or addictive drugs. Sexual stimuli are also powerful appetitive USs.



Aversive Conditioning

In aversive conditioning, the US is an event that an organism generally avoids. In mentalistic terms, it is something typically regarded as unpleasant.

- Examples: An electric shock, a painful bite, or an unpleasant odor.

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Conditioning of Appetitive Responses

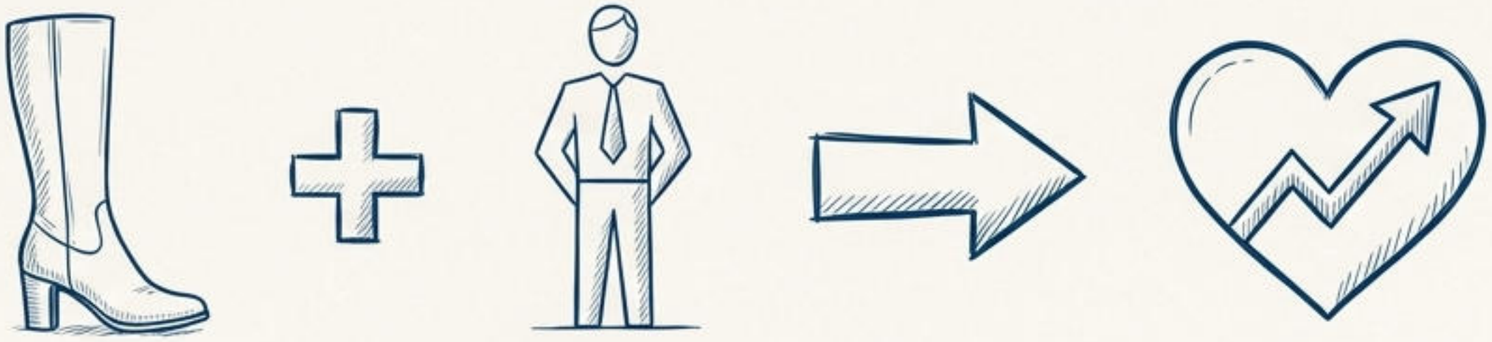
Sexual responses, which are powerfully appetitive, can be effectively shaped through classical conditioning. This has been demonstrated in controlled laboratory settings.

A Case Study: Rachman and Hodgson (1968)

This experiment provides clear evidence for the conditioning of sexual arousal.

- **Participants:** Seven male volunteers.
- **Procedure:** A neutral stimulus (a picture of black, knee-length boots) was consistently followed by an unconditioned stimulus (a picture of a nude woman).
- **Conditioning Trials:** After approximately 30 pairings of the boots (NS) and the nude picture (US).
- **Result:** Five of the seven participants began to exhibit sexual arousal (CR) in response to the picture of the boots alone (now a CS).

(Note: The researchers later eliminated this conditioned response through extinction, by repeatedly presenting the picture of the boots without the subsequent picture of the nude woman.)



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The Power of Aversive Conditioning

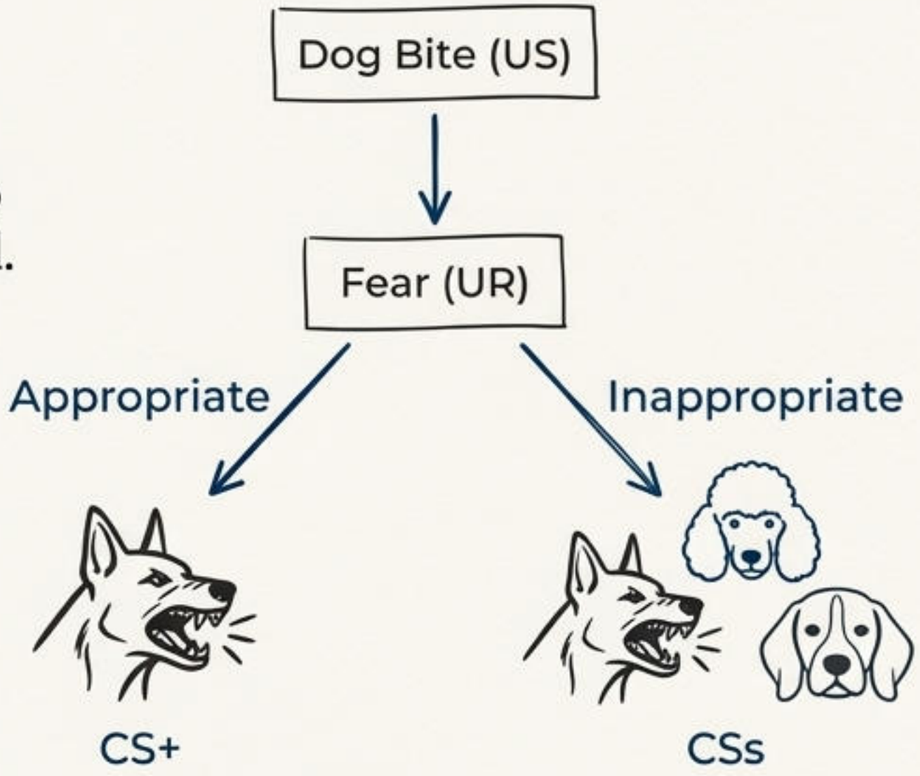
Aversive conditioning often occurs with remarkable speed, sometimes requiring only one or two pairings of the NS and a strong US. This rapid learning reflects a fundamental survival mechanism.

Evolutionary and Survival Relevance

We have evolved to quickly learn associations with events that cause pain, illness, or threat. This rapid conditioning helps us to avoid danger in the future, thus playing a critical role in survival. This process underlies many of our fears and anxieties.

Appropriate vs. Inappropriate Fear Learning

- **Beneficial Fear:** When the conditioning is appropriate and adaptive. For example, learning to fear a specific vicious dog that has bitten you is beneficial for future safety.
- **Problematic Fear:** When the conditioning is inappropriate or overgeneralized. For example, developing a fear of *all* dogs after being bitten by one can become problematic and maladaptive.



This distinction is why significant research has focused on understanding fear conditioning and developing methods to treat inappropriate fears.

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A Challenge for Researchers: Measuring Fear

Given the importance of aversive conditioning, a key challenge emerged: how can the covert emotional response of "fear" be reliably and objectively measured in animal subjects? Direct physiological measures like heart rate can be difficult to record, especially in small animals like rats.

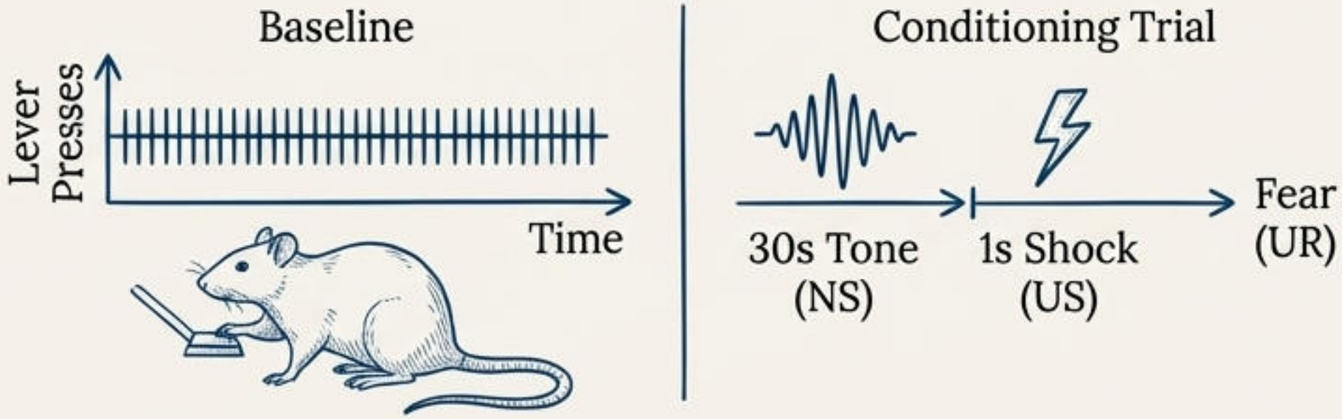
The CER Paradigm: An Ingenious Solution

Estes and Skinner (1941) developed a method known as the **conditioned suppression** or **Conditioned Emotional Response (CER) paradigm**. This provides an indirect but powerful measure of fear.

The Two-Phase Procedure

1. **Baseline Operant Behavior:** First, a rat is trained to engage in an ongoing behavior, such as pressing a lever to obtain food. This establishes a steady, predictable rate of response.
2. **Fear Conditioning:** A classical conditioning procedure is then introduced over the baseline behavior. For example, a 30-second tone (NS) is presented, immediately followed by a brief 1-second shock (US).

`30s Tone (NS) : 1s Shock (US) → Fear (UR)`



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How Suppression Reveals Conditioned Fear

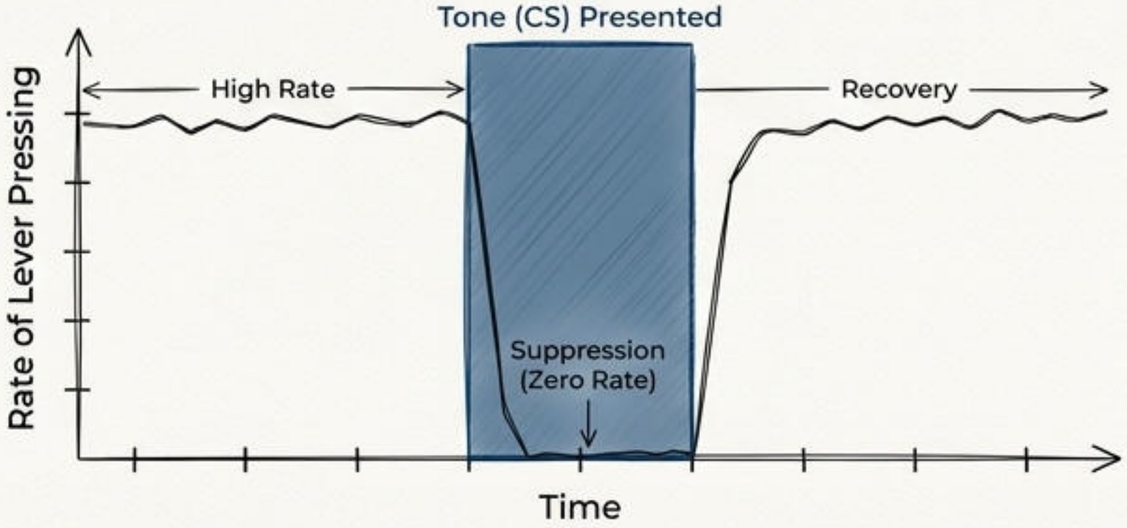
As the conditioning trials proceed, the tone (NS) becomes associated with the shock (US). Consequently, the tone itself begins to elicit the emotional response of fear, transforming it into a conditioned stimulus (CS).

`30s Tone (CS) → Fear (CR)`

An Indirect Measure of Fear

The rat initially stops pressing the lever only when it receives a shock. After conditioning, however, it freezes and stops pressing the lever as soon as it hears the tone. Therefore, the degree to which lever pressing is suppressed during the presentation of the tone serves as an indirect measure of the fear that the tone elicits.

Think of it like patrons in a saloon who stop talking whenever a feared gunfighter enters. The level of silence (suppression of talking) is a precise measure of their fear. The rat's suppression of lever pressing works the same way.



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Quantifying Fear: The Suppression Ratio

Conditioned suppression is measured formally using a suppression ratio. This calculation compares the rate of responding during the CS with the rate of responding just before the CS.

The Suppression Ratio Formula

The formula is the number of responses emitted during the CS period, divided by the total number of responses emitted during the CS period and the same length period immediately preceding the CS.

$$\text{Suppression Ratio} = \frac{\text{\# of CS responses}}{(\text{\# of CS responses} + \text{\# of pre-CS responses})}$$

- **# of CS responses**: Responses during the tone.
- **# of pre-CS responses:** Responses in the period of equal duration just before the tone.

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Interpreting the Suppression Ratio

The value of the suppression ratio indicates the strength of the conditioned fear.

Example Calculations

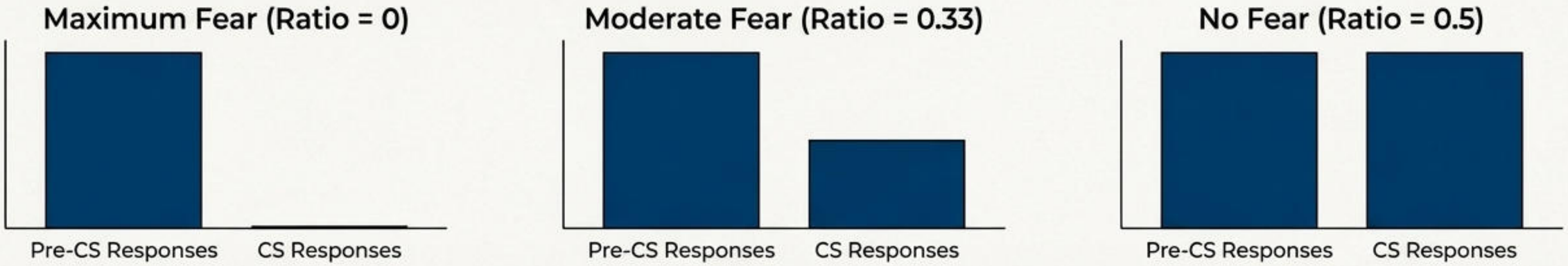
Imagine a rat makes 20 responses in the 30-second pre-CS period.

- **Total Suppression (Maximum Fear):** The rat makes 0 responses during the 30-second CS. Ratio = 0 / (0 + 20) = **0**
- **Partial Suppression (Moderate Fear):** The rat makes 10 responses during the CS. Ratio = 10 / (10 + 20) = 10 / 30 = **0.33**
- **No Suppression (No Fear):** The rat makes 20 responses during the CS, same as before. Ratio = 20 / (20 + 20) = 20 / 40 = **0.5**

The Counterintuitive Scale: 0 to 0.5

It is crucial to remember that the ratio generally varies between 0 and 0.5. The relationship is inverse: **a lower ratio indicates greater suppression** and therefore **stronger fear conditioning**.

Mnemonic: Lower Ratio = Less Responding = Greater Suppression



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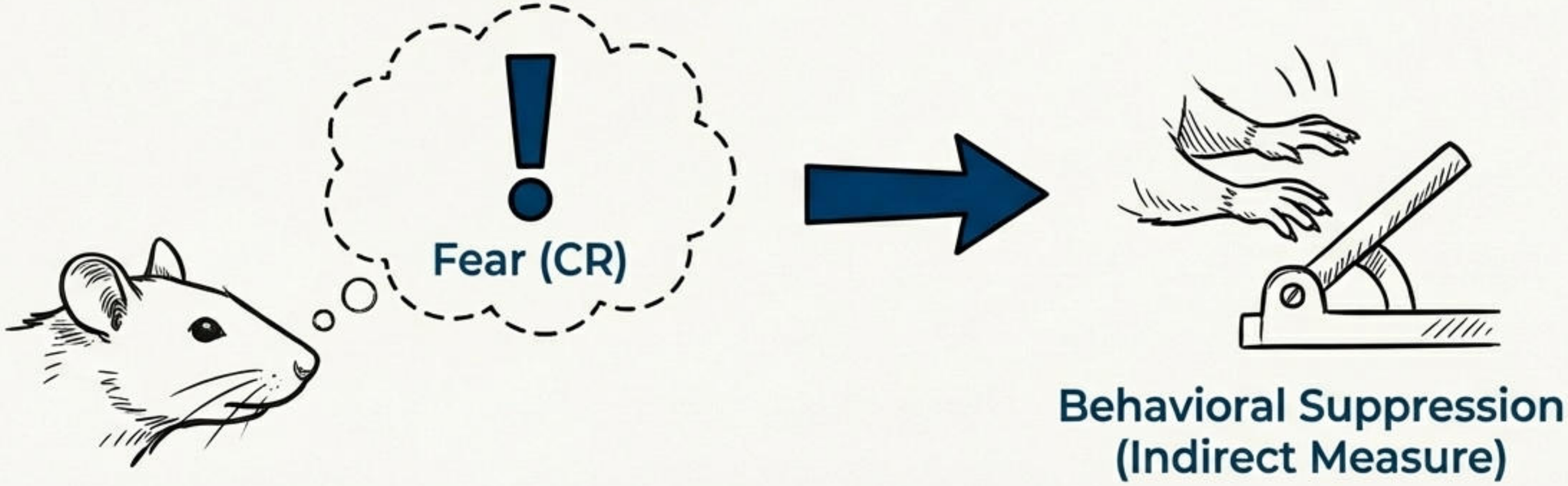
A Critical Clarification: The True Conditioned Response

The CER paradigm is a foundational method used to investigate fear conditioning and other classical conditioning processes. However, a precise conceptual understanding is critical for exam purposes.

The Covert Response of Fear

Remember that the Conditioned Response (CR) in this procedure is the covert, internal response of fear. The reduction in lever pressing is not the CR. It is the observable, indirect measure that allows us to infer the existence and strength of the unobservable, internal fear response.



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Excitatory and Inhibitory Conditioning

Classical conditioning is not limited to creating responses; it can also prevent them. This introduces two fundamental types of conditioned stimuli: excitatory and inhibitory.

Excitatory Conditioning (CS+)

This is conditioning where the Neutral Stimulus (NS) is associated with the presentation of a US. This is the standard.

- Result: The CS (labeled a CS+) comes to elicit a conditioned response.
- Example: A tone (CS+) is paired with shock and comes to elicit fear.

Inhibitory Conditioning (CS–)

This is conditioning where the NS is associated with the absence or removal of a US.

- Result: The CS (labeled a CS–) comes to inhibit the occurrence of a conditioned response. The response is less likely to occur when the CS– is present.
- Example: A light signals that no shock will occur, thereby inhibiting fear.

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Inhibitory Conditioning in Action: Safety Signals

A CS− effectively becomes a "safety signal," communicating that an aversive US will not occur.

Real-World Example

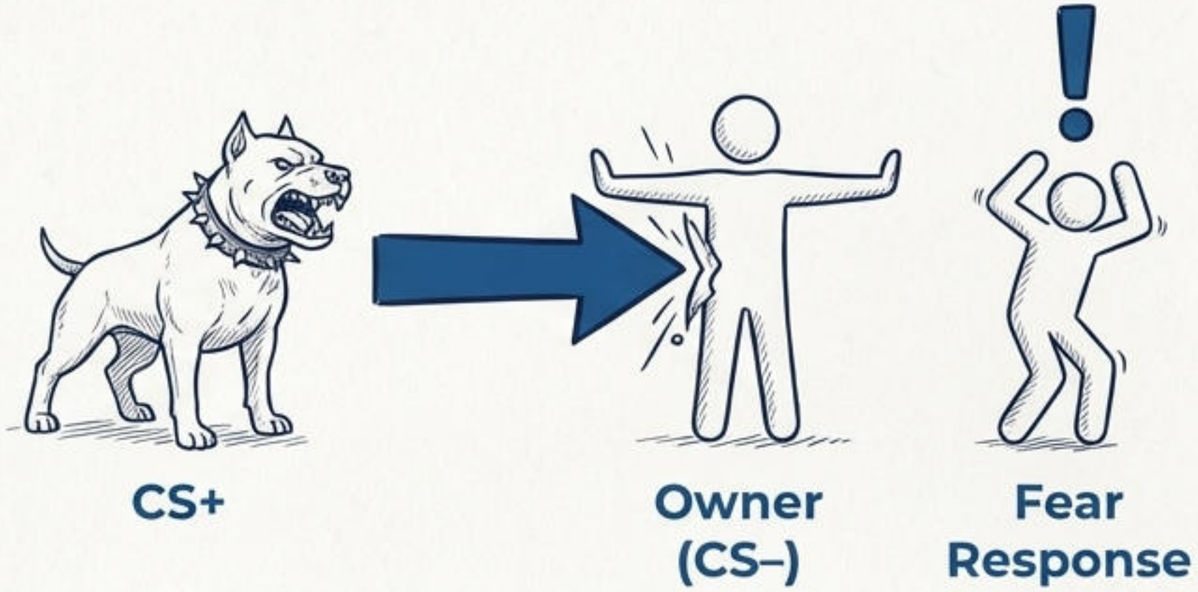
Imagine a vicious dog (which has bitten you) is an excitatory CS+ for fear. However, the dog never bites you when its owner is present.

The dog (CS+): Elicits fear.
The owner (CS−): Becomes a safety signal that predicts the absence of a bite. Your fear is therefore inhibited, or suppressed, when the owner is present.

Experimental Example

A rat is consistently shocked following a tone, making the tone a CS+ for fear. However, if the tone and a light are presented together, no shock ever occurs.

The tone (CS+): Elicits fear.
The light (CS−): Becomes an inhibitory stimulus that signals the absence of shock, thereby inhibiting the fear response elicited by the tone.



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Temporal Arrangement of Stimuli

The effectiveness of classical conditioning is profoundly influenced by the timing between the NS and the US. There are four primary ways to arrange these stimuli.

1. Delayed Conditioning

- **Definition**: The onset of the NS precedes the onset of the US, and the two stimuli overlap. The NS is still "on" when the US begins.
- **Effectiveness**: This is often the most effective arrangement for producing strong conditioning.
- **Interstimulus Interval (ISI):** This is the time between the *onset* of the NS and the *onset* of the US. Shorter ISIs are typically optimal.
  - **Autonomic Responses** (e.g., salivation): Optimal ISI is a few seconds.
  - **Skeletal Responses** (e.g., eyeblink): Optimal ISI is about half a second.

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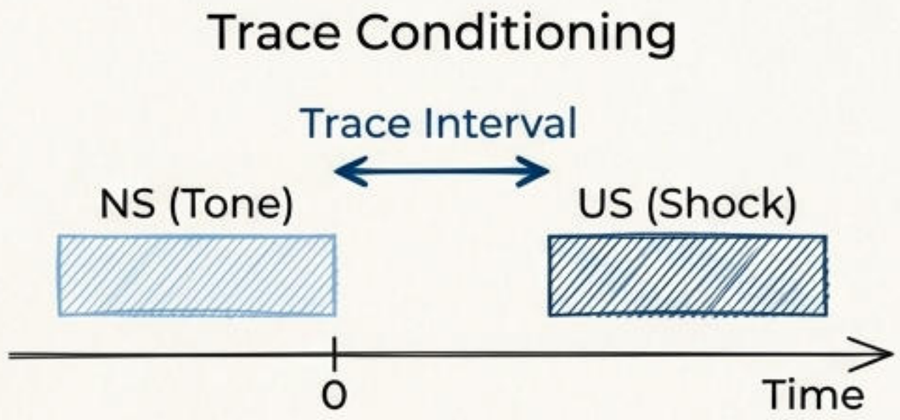
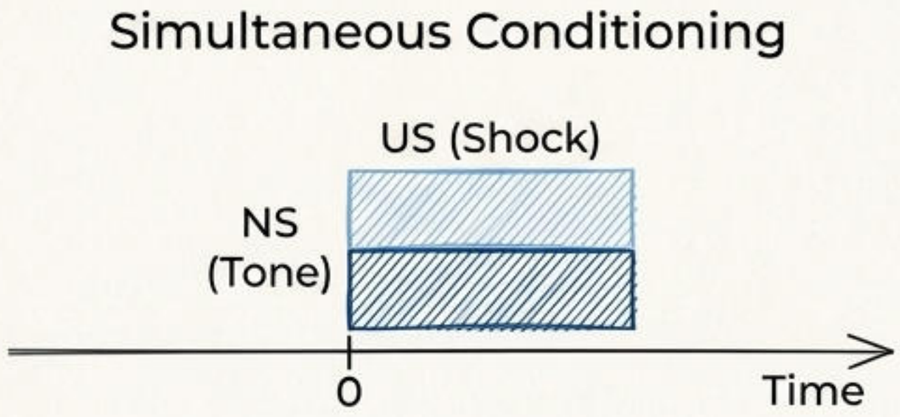
Other Temporal Arrangements

2. Trace Conditioning
   - **Definition:** The onset and offset of the NS precede the onset of the US. The two stimuli do not overlap. The organism must rely on a “memory trace” of the NS.
   - **Trace Interval:** The period between the end of the NS and the start of the US.
   - **Effectiveness:** Can be very effective if the trace interval is short (no more than a few seconds). Effectiveness declines as the interval lengthens.

3. Simultaneous Conditioning
   - **Definition:** The onset of the NS and the onset of the US occur at the exact same time.
   - **Effectiveness:** This procedure usually results in poor conditioning. Because the NS does not precede the US, it does not serve as a useful predictor.





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Temporal Arrangement of Stimuli (continued)

4. Backward Conditioning

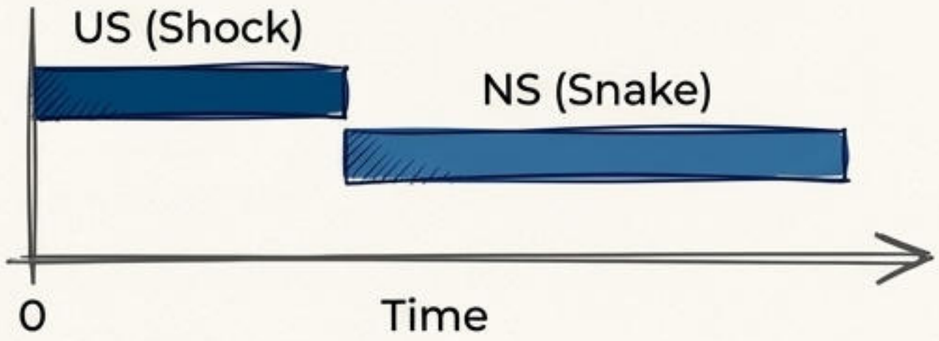
- **Definition:** The onset of the NS *follows* the onset of the US. The US is presented first, and the NS is presented later.
- **Effectiveness:** This is traditionally considered the least effective procedure for producing **excitatory** conditioning.

An Exception: Biological Preparedness

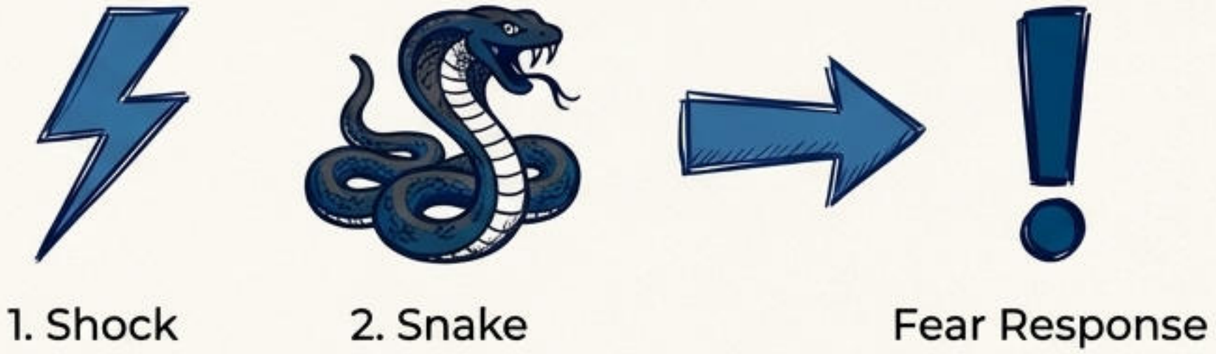
Under specific circumstances, backward excitatory conditioning can occur. This is often observed when the NS is a stimulus for which an organism has an innate predisposition to fear.

- **Example:** For a rat, a shock (US) followed by the sight of a snake (NS) may still produce a conditioned fear of the snake.
- **Reasoning:** Some researchers propose that species have an inherited predisposition to fear evolutionarily relevant threats (e.g., rats fearing snakes). This biological relevance is so strong that an association can form even with a suboptimal temporal arrangement.

Backward Conditioning



Biological Preparedness



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The Dual Role of Backward Conditioning

While often ineffective for excitatory conditioning, the backward arrangement can be highly effective for producing inhibitory conditioning.

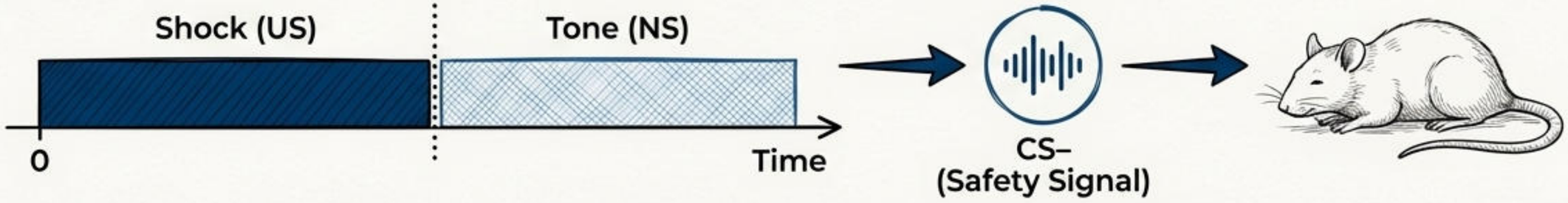
Backward Conditioning as Inhibitory

This occurs when the NS signals the removal or termination of an aversive US.

- **Lab Example:** A tone sounds just as a shock is turned off. The tone reliably predicts the end of the shock and thus becomes a safety signal (CS–) that inhibits fear.
- **Real-World Example:** A child suffering from an asthma attack (aversive US) feels relief when a doctor administers an injection. The doctor's presence can become a CS- that signals safety and inhibits the child's distress.

Concluding Note

While delayed conditioning is often the most robust arrangement, it is clear that all temporal arrangements can influence behavior. Conditioning is not merely about contiguity; it is critically about the predictive information that a stimulus provides.



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